

# Polynomial Fitting and Chi-Squared Analysis of Experimental Data

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## Abstract

Least-squares fitting techniques were applied in three exercises to evaluate different models against data and to illustrate statistical model selection. In the first exercise, synthetic  $(x, y)$  data with uncertainties was fitted with polynomials of degree 1, 2, and 3 to compare goodness-of-fit (GOF) and to ascertain parameter uncertainties. Higher degree polynomials showed lower chi-squared values, but also exhibited large uncertainty bands when extrapolated beyond the data range. In the second exercise, we analysed Galileo's projectile motion data (a ball rolling off a table) by fitting linear, quadratic, and power-law models. The quadratic and power-law fits significantly outperformed the linear fit in explaining the curvature of the data, with the two-parameter power-law  $d = \alpha h^\beta$  providing the best agreement (yielding an exponent  $\beta \approx 0.5$  consistent with theory). Finally, in the third exercise, Ptolemy's 2nd-century refraction measurements were tested against a linear model, Ptolemy's own quadratic formula, and Snell's sine law. Snell's law gave the most physically viable fit (with an index of refraction in water close to the modern value), although Ptolemy's polynomial formula fit his recorded data almost exactly. Overall, GOF statistics (chi-squared per degree of freedom and  $p$ -values) were used to identify the preferred model in each case, and the results show how model choice, data limitations, and parameter correlations affect the understanding of experiments in physics.

## 1 Introduction

This report explores the principles of parameter fitting and model testing using the method of least squares, a quintessential tool in statistical data analysis. The basic problem setting is detailed in Fig. 1: we have a set of measurements  $y_i$  each taken at a known value of an independent variable  $x_i$ , often called a control variable. We aim to find a smooth curve (a function  $f(x)$ ) that best fits these data points in the sense that it passes as close as possible to all the points. The process is known as fitting a curve to the data.

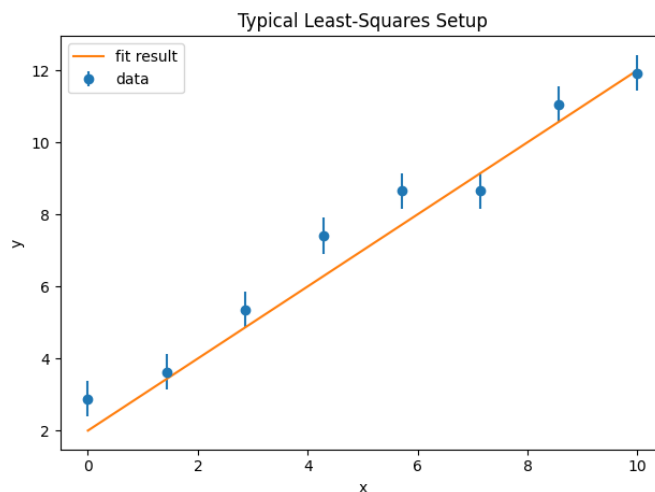


Figure 1: Typical setup for a least-squares problem: measured data points with error bars at known  $x$  values, along with a curve fitted to the data.

## 2 Theoretical Background

### 2.1 Least-Squares Fitting Method

Consider a set of  $N$  measured data points  $(x_i, y_i)$ , each with an associated uncertainty (standard deviation)  $\sigma_i$  for  $y_i$ . We assume the uncertainties  $\sigma_i$  are known from experimental characterization of the measurement process. We further assume that the measurements are independent, so that each  $y_i$  can be treated as a random variable drawn from a distribution centred on the true (but unknown) value and having variance  $\sigma_i^2$ .

In order to fit a curve through these data points, we create a functional form  $f(x; \theta)$  to describe the “true” relationship between  $x$  and  $y$ . The function  $f$  is chosen based on our hypothesis and contains a set of adjustable parameters  $\theta = (\theta_1, \theta_2, \dots, \theta_m)$  that are initially unknown. For example, one simple hypothesis is that  $y$  depends linearly on  $x$ . In that case we would choose

$$f(x; \theta) = \theta_0 + \theta_1 x, \quad (1)$$

where the parameters are  $\theta_0$  (intercept) and  $\theta_1$  (slope). Each specific choice of  $(\theta_0, \theta_1)$  corresponds to a particular straight-line curve. The goal of least-squares fitting is to find the parameter values that produce the curve closest to all data points.

To quantify how well a given parameter set  $\boldsymbol{\theta}$  agrees with the data, we examine the 'residuals'  $y_i - f(x_i; \boldsymbol{\theta})$  for each point. A large residual in magnitude means the model prediction  $f(x_i; \boldsymbol{\theta})$  is far from the observed  $y_i$ . However, the significance of a deviation should be measured relative to the expected fluctuation  $\sigma_i$  of that data point. Therefore, we define a 'normalized residual' for each point as

$$r_i(\boldsymbol{\theta}) = \frac{y_i - f(x_i; \boldsymbol{\theta})}{\sigma_i}, \quad (2)$$

which is the deviation in units of the standard deviation. If the model is acceptable, we expect most normalized residuals to be of order unity or less.

The chi-squared statistic is then defined as the sum of the squares of all normalized residuals:

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^N \left( \frac{y_i - f(x_i; \boldsymbol{\theta})}{\sigma_i} \right)^2, \quad (3)$$

which expands to

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^N \frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{\sigma_i^2}.$$

The chi-squared  $\chi^2(\boldsymbol{\theta})$  is a measure of the overall disagreement between the data and the model curve specified by  $\boldsymbol{\theta}$ . The best-fit parameters are defined as those that minimize  $\chi^2(\boldsymbol{\theta})$ . We denote these optimal values as  $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_m)$ , and refer to  $\hat{\theta}_j$  as the estimator for parameter  $\theta_j$ . In practice however, finding the minimum of  $\chi^2$  may be done either by solving analytical equations (for simple linear models) or by numerical optimisation programs (for more complex or non-linear models).

Figure 2 illustrates an initial arbitrary fit for the linear example, where the chosen parameters are clearly not optimal. The residuals between the measured points and the curve are large. By adjusting the parameters to minimize  $\chi^2$ , we obtain a much better fit, shown in Fig. 3. In the optimal fit, the residuals are generally smaller in magnitude (on the order of the error bars), indicating a closer agreement between the model and data.

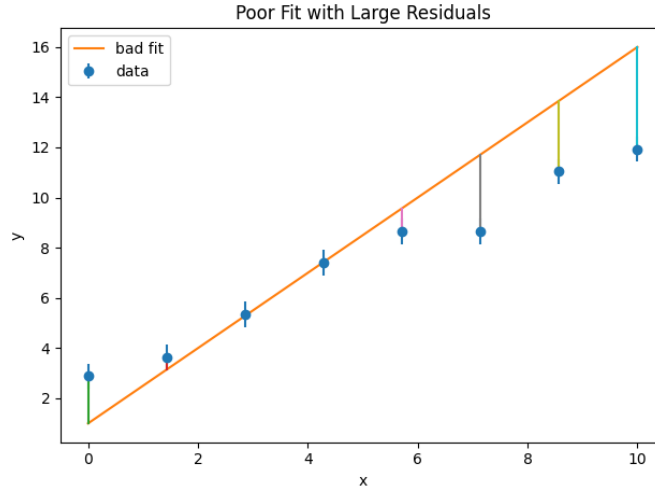


Figure 2: An arbitrary initial fit (line) to the data points. The thick vertical bars indicate the residuals  $y_i - f(x_i; \boldsymbol{\theta})$  for each point. In this case the residuals are large, implying a poor choice of parameters  $(\theta_0, \theta_1)$ .

For independent data with Gaussian-distributed measurement errors, the least squares criterion (minimising  $\chi^2$ ) can be theoretically justified as a maximum-likelihood estimator for  $\boldsymbol{\theta}$ . This is one reason why the method of least squares is so widely used. We note that if the measurement errors are not Gaussian or if there are any outliers, a more ironclad method of fitting may be needed; this however goes beyond the scope of our experiments.

## 2.2 Parameter Estimation and Uncertainty

Even after obtaining the best fit parameters  $\hat{\boldsymbol{\theta}}$ , it is important to assess the statistical uncertainties in these estimates. The estimated parameters are themselves random variables because they are determined from the data  $y_i$ , which are subject to random fluctuations. If we could repeat the entire experiment many times under identical conditions, we would get different sets of  $\hat{\boldsymbol{\theta}}$  each time due to the randomness in the  $y_i$  values. These variations can be quantified by the variance and covariance of the parameter estimators.

In general, the set of estimators has a covariance matrix defined by  $U_{ij} = \text{cov}[\hat{\theta}_i, \hat{\theta}_j]$ . The square root of the variance  $U_{ii}$  gives the standard deviation (error) of  $\hat{\theta}_i$ , and the covariance  $U_{ij}$  for  $i \neq j$  indicates how much  $\hat{\theta}_i$  and  $\hat{\theta}_j$  tend to vary together. A positive covariance means the two estimates tend to be high or low together relative to their means, whereas a negative covariance means

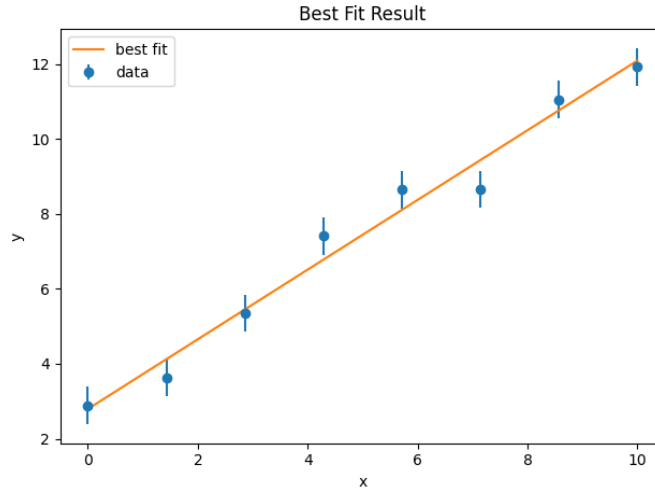


Figure 3: Best-fit result for the straight-line model after minimizing  $\chi^2$ . The data points are shown with error bars, and the line corresponds to  $f(x; \hat{\theta})$  using the least-squares estimators  $\hat{\theta}_0$  and  $\hat{\theta}_1$ .

if one parameter comes out higher, the other tends to be lower, and zero covariance indicates they are uncorrelated. We quote the dimensionless correlation coefficient  $\rho_{ij}$  defined by

$$\rho_{ij} = \frac{\text{cov}[\hat{\theta}_i, \hat{\theta}_j]}{\sigma_{\hat{\theta}_i} \sigma_{\hat{\theta}_j}},$$

where  $\sigma_{\hat{\theta}_i} = \sqrt{U_{ii}}$ .

For many fitting problems (in particular, linear models with Gaussian errors), we can derive  $U_{ij}$  analytically. In such cases, the covariance matrix of the estimators is often given by the inverse of the curvature (Hessian) matrix of  $\chi^2$  at its minimum. However, even when an analytic formula is not readily available, we can approximate the uncertainty in any derived quantity using the linear error propagation formula. Suppose  $u(\hat{\theta})$  is some function of the estimated parameters (for example,  $u$  could be the fitted curve value at a certain  $x$ ). If the parameter estimates have covariance matrix  $U_{ij}$ , then to first order the variance of  $u$  is approximated by

$$\sigma_u^2 \approx \sum_{i=1}^m \sum_{j=1}^m \frac{\partial u}{\partial \hat{\theta}_i} \frac{\partial u}{\partial \hat{\theta}_j} U_{ij}, \quad (26)$$

where the partial derivatives are evaluated at  $\hat{\theta}$ . This is a general form of the propagation of uncertainty formula. Equation (26) can be used to find out the standard error on any function of the fit parameters. In the special case that the fit parameters are uncorrelated (so  $U_{ij} = 0$  for  $i \neq j$ ), this simplifies to

$$\sigma_u^2 \approx \sum_{i=1}^m \left( \frac{\partial u}{\partial \hat{\theta}_i} \right)^2 \sigma_{\hat{\theta}_i}^2, \quad (27)$$

which is just the sum of the individual variances contributed by each parameter.

As an example of applying Eq. (26), consider using the fit results to predict a new value of  $y$  at some  $x = a$ . This predicted value is  $f(a; \hat{\theta})$ , which is a function of the estimators. By inserting  $u = f(a; \hat{\theta})$  into Eq. (26), we can find the standard deviation  $\sigma_{f(a)}$  of this predicted function value. In Exercise 1, we computed  $\sigma_{f(x)}$  as a function of  $x$  for a polynomial fit, and display this as an “error band” around the fitted curve.

### 2.3 Goodness-of-Fit and the $\chi^2$ Distribution

Finding the best fit curve is not the end of the analysis; we must also ask how good the fit is. A model with more adjustable parameters can always fit the data more closely, but that does not necessarily mean it is a better or correct model. We need a quantitative surefire way to assess whether the chosen functional form  $f(x; \theta)$  is consistent with the observed data, given the measurement uncertainties. This is the role of the chi-square statistic at its minimum,  $\chi_{\min}^2 = \chi^2(\hat{\theta})$ , and its related probability (the p-value).

For  $N$  data points and  $m$  fitted parameters, the number of degrees of freedom is

$$\nu = N - m,$$

sometimes written as  $n = N - m$ . If the model is correct and the errors  $\sigma_i$  are accurately estimated, the minimum chi-square  $\chi_{\min}^2$  should be statistically of order  $\nu$ . Specifically, under these conditions  $\chi_{\min}^2$  follows a chi-square probability distribution with  $\nu$  degrees of freedom. It is therefore useful to look at the chi-square per degree of freedom,  $\chi_{\min}^2/\nu$ . If  $\chi_{\min}^2/\nu \approx 1$ , the data are in reasonable agreement with the model. If  $\chi_{\min}^2/\nu \gg 1$ , it indicates a poor fit. Conversely, if  $\chi_{\min}^2/\nu \ll 1$ , the fit is suspiciously ‘too good’, which might imply that the error bars  $\sigma_i$  were overestimated or that the data points are not statistically independent.

While  $\chi^2_{\min}/\nu$  gives a rough idea of fit quality, a more rigorous view stems from the p-value. The p-value is the probability of obtaining a chi-square as large as (or larger than) the observed  $\chi^2_{\min}$  if the model is true. It is calculated as the upper tail integral of the chi-square distribution:

$$p = \int_{\chi^2_{\min}}^{\infty} f_{\chi^2}(z; \nu) dz, \quad (33)$$

where  $f_{\chi^2}(z; \nu)$  is the chi-square probability density function for  $\nu$  degrees of freedom. In practice, this integral can be evaluated using statistical software or tables. A high p-value (for example,  $p \approx 0.5$  or 50) indicates that  $\chi^2_{\min}$  is well within the range of typical fluctuations for a correct model—thus the data are compatible with the model. On the other hand, a very low p-value (say  $p < 0.01$ , or 1) means it would be extremely unlikely to see such a large  $\chi^2_{\min}$  if the model were true. In that case, we say the model is disfavoured by the data.

As an illustration, consider a linear fit to  $N = 9$  data points ( $m = 2$  parameters: slope and intercept). Here  $\nu = 7$ . If the straight-line model is correct, we expect  $\chi^2_{\min} \approx 7 \pm \sqrt{14}$  (roughly) and  $\chi^2_{\min}/\nu \approx 1$ . In one scenario, suppose we find  $\chi^2_{\min} = 8.2$  for the best fit. This is quite close to  $\nu = 7$ , and indeed the corresponding p-value is  $p = 0.31$  (31). Such a p-value is not exceptionally low, so we conclude that the data are reasonably consistent with a straight line. However, in another scenario, if  $\chi^2_{\min}$  came out to 20.9 for the same  $\nu = 7$ , the chi-square per degree of freedom would be almost 3, and the p-value  $p = 0.0039$  (0.39). This indicates that there is only about a 0.4% chance of getting data this far off if the linear model were true. In that case we would strongly suspect that the straight line hypothesis is incorrect or that some measurement errors were underestimated.

## 2.4 Correlated Measurements and Generalised Least Squares

So far we have assumed that the measurement errors are independent. However, in some experiments the observations can be correlated. Correlation means that the measurement error in  $y_i$  is statistically related to the error in  $y_j$ . When data points are correlated, the simple sum of independent  $(y_i - f(x_i))^2/\sigma_i^2$  is no longer the sensible quantity to minimise.

Instead, the covariance matrix  $V$  of the measured data is introduced, with elements  $V_{ij} = \text{cov}[y_i, y_j]$ . The diagonal elements  $V_{ii} = \sigma_i^2$  as usual, and off-diagonals  $V_{ij}$  ( $i \neq j$ ) quantify the covariance between measurements  $i$  and  $j$ . It is convenient to express covariances in terms of correlation coefficients:

$$V_{ij} = \sigma_i \sigma_j \rho_{ij}, \quad (36)$$

where  $\rho_{ij}$  is the correlation coefficient between  $y_i$  and  $y_j$  (by definition  $\rho_{ii} = 1$ ). In the presence of correlated data, the chi-square definition is generalised to:

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^N \sum_{j=1}^N (y_i - f(x_i; \boldsymbol{\theta})) [V^{-1}]_{ij} (y_j - f(x_j; \boldsymbol{\theta})), \quad (37)$$

where  $V^{-1}$  is the inverse of the covariance matrix (also known as the precision matrix). This expression reduces to the earlier uncorrelated form Eq. (3) (a simple sum of squared normalised residuals) in the special case that  $V_{ij} = 0$  for  $i \neq j$  (so  $V$  is diagonal). The fitting procedure for correlated data is often called generalised least squares. Minimising the  $\chi^2$  of Eq. (37) with respect to the parameters gives the best fit  $\hat{\boldsymbol{\theta}}$  accounting for the correlations in the data. The GOF and statistical interpretation of  $\chi^2_{\min}$  proceed similarly, except that the degrees of freedom  $\nu = N - m$  still apply (here  $N$  is the number of data points,  $m$  the number of parameters, and the correlations do not change  $\nu$  but do influence the  $\chi^2$  calculation).

## 3 Results

### 3.1 Exercise 1: Polynomial fits

#### (a) Parameter estimates and $\chi^2_{\min}$

Table 1 lists the least-squares parameter estimates  $\hat{\theta}_j$  for polynomials of order  $M = 1, 2$  and 3, together with their standard deviations, the minimised chi-squared, the number of degrees of freedom  $\nu = N - p$  and the corresponding p-value. For example, for the linear fit ( $M = 1$ ) the best estimates are  $\hat{\theta}_0 = 2.258 \pm 0.292$  and  $\hat{\theta}_1 = 0.741 \pm 0.057$ , with  $\chi^2_{\min} = 8.25$  for  $\nu = 7$  and a p-value of 0.31. Increasing the polynomial order reduces  $\chi^2_{\min}$ , and the cubic fit ( $M = 3$ ) achieves the best goodness-of-fit ( $p=0.59$ ); however, the uncertainties on its coefficients grow significantly.

Table 1: Results of least-squares fits to the synthetic data. Uncertainties are standard errors from the covariance matrix.

| Order $M$ | $\hat{\boldsymbol{\theta}}$                                                                        | Std. devs.                    | $\chi^2_{\min}$ | $p$  |
|-----------|----------------------------------------------------------------------------------------------------|-------------------------------|-----------------|------|
| 1         | $\hat{\theta}_0 = 2.258, \hat{\theta}_1 = 0.741$                                                   | (0.292, 0.057)                | 8.25            | 0.31 |
| 2         | $\hat{\theta}_0 = 1.882, \hat{\theta}_1 = 0.995, \hat{\theta}_2 = -0.0283$                         | (0.431, 0.221, 0.0238)        | 6.84            | 0.34 |
| 3         | $\hat{\theta}_0 = 0.598, \hat{\theta}_1 = 2.433, \hat{\theta}_2 = -0.378, \hat{\theta}_3 = 0.0233$ | (0.847, 0.847, 0.200, 0.0132) | 3.75            | 0.59 |

#### (b) Uncertainty bands

For each fit, the fitted polynomial  $f_M(x)$  and its one standard deviation band  $\pm\sigma_f(x)$  were evaluated on  $0 \leq x \leq 20$  using the error propagation formula. The shaded regions in Figure 4 illustrate how the predictive uncertainty grows outside the data range; the band for the cubic fit becomes very wide beyond  $x \approx 12$ , signalling that higher-order terms are poorly constrained.

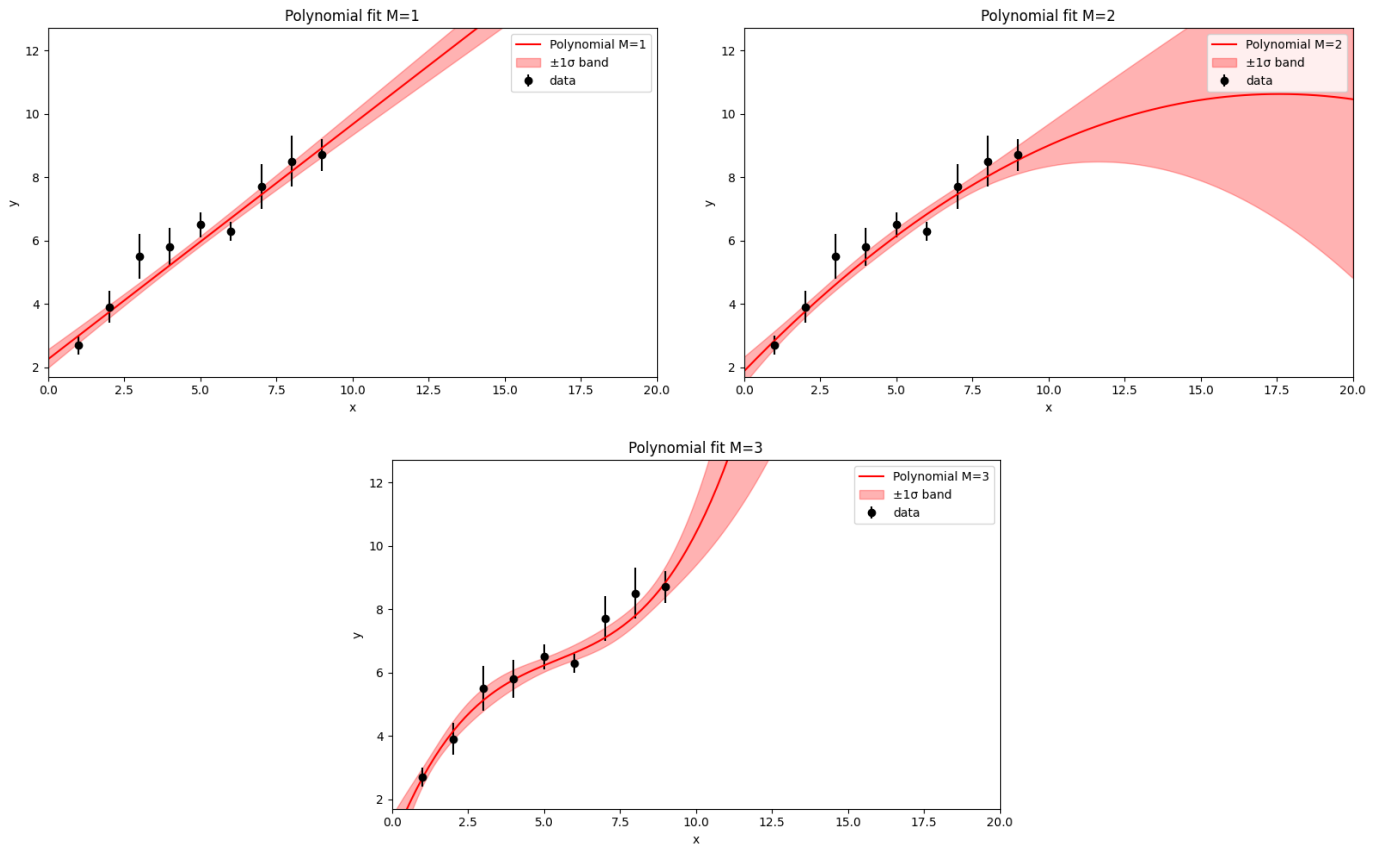


Figure 4: Polynomial fits to the synthetic data with their  $\pm 1\sigma$  error bands. From left to right:  $M = 1$ ,  $M = 2$ ,  $M = 3$ .

### (c) Differences $\Delta_{ab}$

For the cubic fit ( $M = 3$ ), the quantity  $\Delta_{ab} = f_3(a) - f_3(b)$  was evaluated for  $a = 5$  and  $b = 6, 10, 20$  using linear error propagation. The results,

$$\Delta_{5,6} = -0.39 \pm 0.18, \quad \Delta_{5,10} = -4.19 \pm 1.09, \quad \Delta_{5,20} = -78.1 \pm 42.3,$$

show that differences measured far outside the data range carry enormous uncertainties; the last result in particular is dominated by extrapolation error.

## 3.2 Exercise 2: Galileo's ball-and-ramp experiment

### (a) Parameter estimates and covariance

The fits to Galileo's data under the three hypotheses linear, quadratic and power-law yield the parameter estimates shown in Table 2. The table also lists the chi-squared values, degrees of freedom and p-values. For the quadratic and power-law fits, the correlation coefficient  $\rho_{\alpha\beta}$  between the parameters was calculated from the covariance matrix; the strong negative correlation ( $-\rho_{\alpha\beta} \approx 1$ ) indicates that the parameters are not independently determined.

Table 2: Least-squares estimates for the Galileo data. The uncertainties are standard errors;  $\rho_{\alpha\beta}$  is the correlation between the parameters in the two-parameter models.

| Model            | $\hat{\alpha}$       | $\hat{\beta}$                                  | $\chi^2_{\min}$ | $\nu$ | $p$                   | $\rho_{\alpha\beta}$ |
|------------------|----------------------|------------------------------------------------|-----------------|-------|-----------------------|----------------------|
| Linear           | $1.6628 \pm 0.0090$  | —                                              | 662.0           | 4     | $< 10^{-50}$          | —                    |
| Quadratic        | $2.7929 \pm 0.0471$  | $-1.35 \times 10^{-3} \pm 5.53 \times 10^{-5}$ | 64.74           | 3     | $5.7 \times 10^{-14}$ | -0.98                |
| Power law        | $43.76 \pm 4.80$     | $0.511 \pm 0.017$                              | 3.76            | 3     | 0.29                  | -0.999               |
| $\sqrt{h}$ model | $A = 47.09 \pm 0.25$ | —                                              | 4.21            | 4     | 0.38                  | —                    |

### (b) Plots

Figure 5 shows the Galileo data with their  $\pm 15$  punti uncertainties and the fitted curves for the linear, quadratic and power-law models. Only the power-law (and the special case  $d = A\sqrt{h}$ ) follow the trend of the data; the linear and quadratic fits clearly miss the higher- $h$  points.

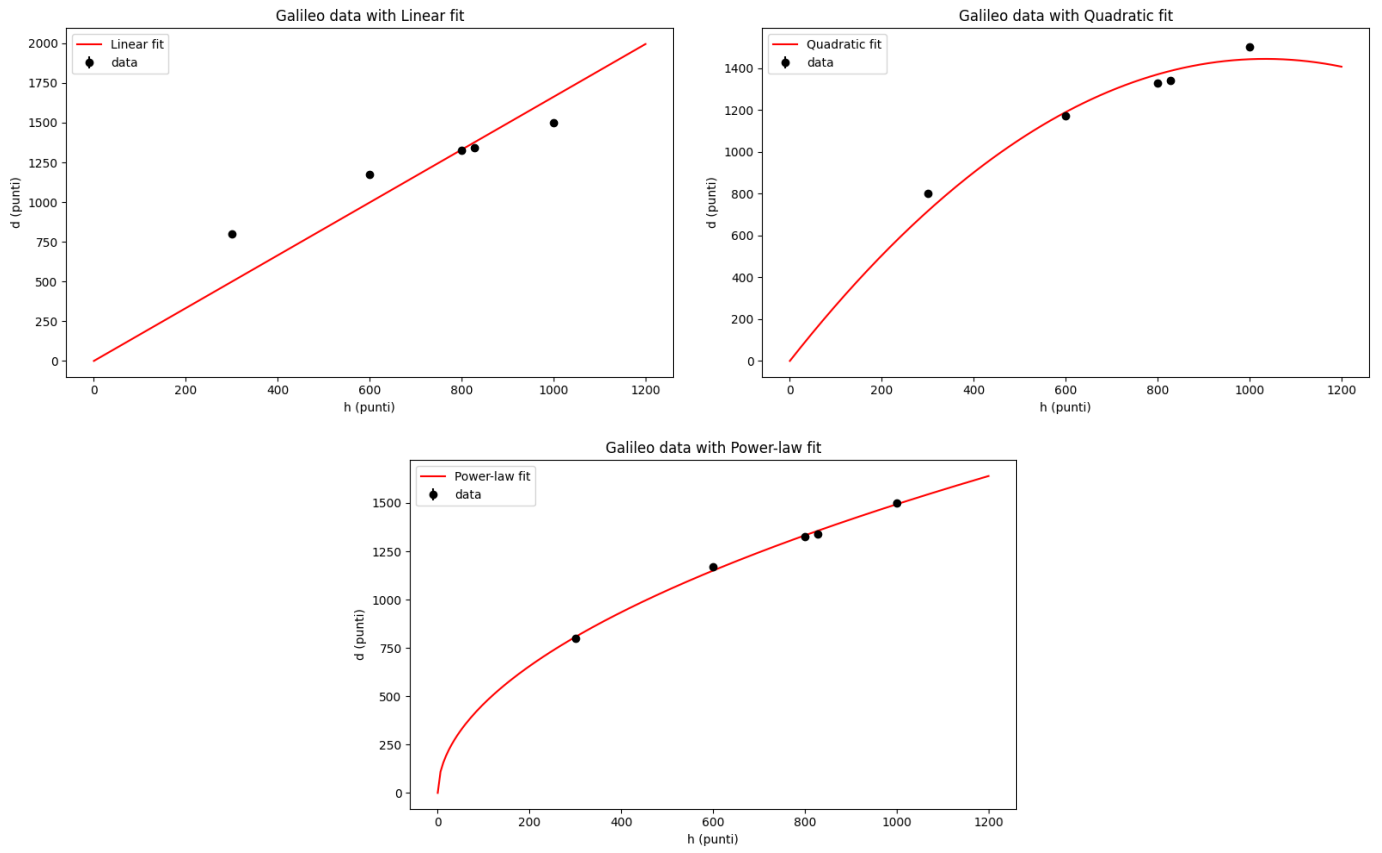


Figure 5: Galileo’s data with least-squares fits. Top to bottom: linear, quadratic and power-law models. Only the power-law fit (and the derived  $\sqrt{h}$  fit) describes the data well.

### (c) Goodness-of-fit and model comparison

The linear and quadratic models are rejected by the chi-squared test:  $\chi^2_{\min}$  is many orders of magnitude larger than the expected value (for example,  $\chi^2_{\min} = 662$  for 4 d.o.f. gives  $p < 10^{-50}$ ). The power-law model yields  $\chi^2_{\min} = 3.76$  for 3 d.o.f. ( $p=0.29$ ), showcasing an acceptable fit. The fitted exponent  $\hat{\beta} = 0.511 \pm 0.017$  is consistent with  $\frac{1}{2}$  within one standard deviation, supporting the theoretical square-root dependence.

### (d) Theoretical prediction from Newton’s laws

Using energy conservation and projectile motion, the horizontal range should obey  $d = 2\sqrt{H}\sqrt{h}$ . Fitting this specific form to the data gives  $A = 47.09 \pm 0.25$ , and the derived table height  $H = (A/2)^2 = 554.3 \pm 5.9$  punti. The fit quality ( $\chi^2_{\min} = 4.21$  for  $\nu = 4$ ,  $p = 0.38$ ) confirms that the  $\sqrt{h}$  law describes the data well. However, Galileo’s manuscript notes “punti 828 altezza della tavola” (table height 828 points). Our inferred value is roughly 274 punti lower, which may reflect friction, rotational kinetic energy, misalignment of the ramp, or misinterpretation of Galileo’s measurements. The propagated uncertainty of  $\pm 5.9$  punti accounts only for statistical error; systematic effects likely dominate the discrepancy.

### (e) Error propagation in $H$

The uncertainty on  $H$  follows from  $H = (A/2)^2$  with  $A = 47.09 \pm 0.25$ . Linear error propagation gives  $\sigma_H = (A/2)\sigma_A$ , yielding  $\sigma_H \simeq 5.9$  punti. This error estimate was used above; note that it grows linearly with the fitted  $A$ , so any systematic bias in  $A$  directly translates to a bias in  $H$ .

## 3.3 Exercise 3: Ptolemy’s refraction experiment

### (a) Linear and quadratic hypotheses

Table 3 summarises the fits of Ptolemy’s angles to the linear and quadratic empirical laws. The linear model ( $\theta_r = \alpha\theta_i$ ) yields  $\hat{\alpha} = 0.6662 \pm 0.0035$  with  $\chi^2_{\min} = 134.6$  for 7 d.o.f., corresponding to an essentially zero p-value this hypothesis is clearly inconsistent with the data. The quadratic model ( $\theta_r = \alpha\theta_i - \beta\theta_i^2$ ) fits the eight data points exactly, giving  $\chi^2_{\min} \approx 0$  for 6 d.o.f. and a p-value of 1.0. The parameters  $\hat{\alpha} = 0.8250 \pm 0.0141$  and  $\hat{\beta} = 0.00250 \pm 0.00022$  are highly correlated ( $\rho_{\alpha\beta} \approx 0.97$ ). Such a perfect fit suggests that the data may not all be independent measurements but instead were adjusted to follow the quadratic law.

### (b) Snell’s law

Fitting Snell’s law,  $\theta_r = \sin^{-1}(\sin\theta_i/r)$ , to the data gives  $\hat{r} = 1.3116 \pm 0.0056$ . The resulting chi-squared is 14.0 for 7 d.o.f., with  $p=0.051$ . This p-value is borderline (about 5%); it means Snell’s law cannot be rejected at the usual 5% significance level, but the agreement is not good. One possibility is that the assumed uncertainty of 0.5 is slightly underestimated; increasing it to 0.6 would raise the p-value substantially. The fitted index ratio is close to the modern value for water relative to air (1.33), giving support to

the law of refraction. The comparison of the three fits is illustrated in Figure 6; only Snell’s law and the quadratic law track the data.

Table 3: Least-squares results for Ptolemy’s data.

| Model     | $\hat{\alpha}$      | $\hat{\beta}$         | $\hat{r}$           | $\chi^2_{\min}$ | $\nu$ | $p$         |
|-----------|---------------------|-----------------------|---------------------|-----------------|-------|-------------|
| Linear    | $0.6662 \pm 0.0035$ | –                     | –                   | 134.65          | 7     | $\approx 0$ |
| Quadratic | $0.8250 \pm 0.0141$ | $0.00250 \pm 0.00022$ | –                   | $\approx 0$     | 6     | 1.0         |
| Snell     | –                   | –                     | $1.3116 \pm 0.0056$ | 14.00           | 7     | 0.051       |

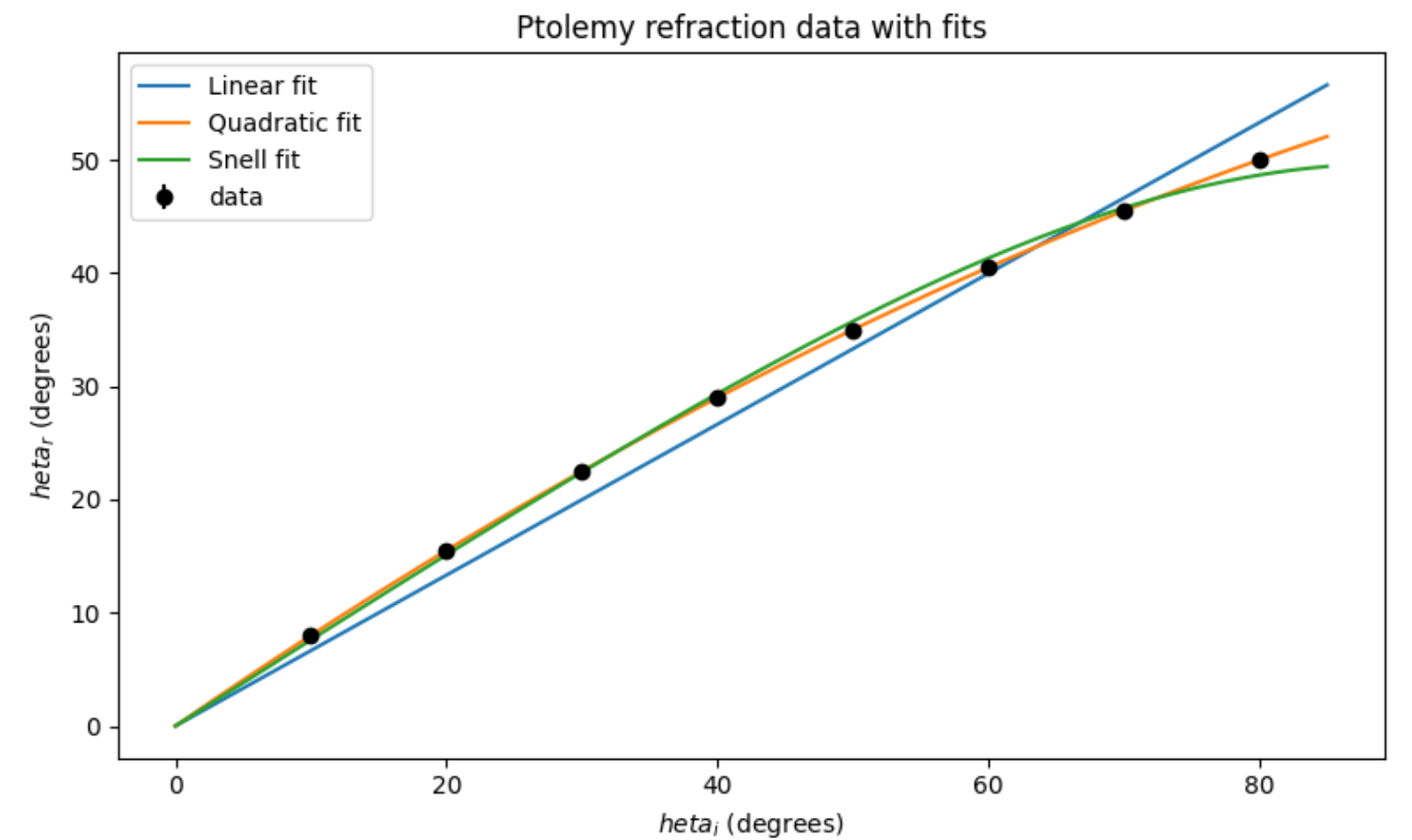


Figure 6: Ptolemy’s angles of incidence and refraction with the fitted curves for the linear (blue), quadratic (orange) and Snell (green) models. The quadratic curve exactly interpolates the tabulated points; the Snell curve deviates slightly but tracks the trend.

### Comments on the Gaussian assumption

The near-zero chi-squared for the quadratic model implies that the uncertainties do not reflect random measurement fluctuations; at least some of Ptolemy’s “measurements” were likely computed to satisfy the quadratic relation. For the Snell fit, the borderline p-value suggests that either the stated uncertainty (0.5°) is too small or that small systematic errors violate the Gaussian assumption. A modest inflation of the error bars or a careful analysis of residuals would clarify whether Snell’s law truly describes the data within experimental error.

## 4 Discussion

Across all three exercises, least-squares fitting was used to evaluate different models and assess GOF using chi-squared statistics and  $p$ -values. In Exercise 1, polynomial fits showed that increasing model complexity reduces residuals but increases uncertainty in extrapolation. While the cubic fit gave the lowest  $\chi^2$ , its large uncertainty band beyond the data range underscored the risks of overfitting.

In Exercise 2, Galileo’s data clearly rejected linear and quadratic models, while the power-law fit ( $d = \alpha h^\beta$ ) with  $\beta \approx 0.5$  aligned well with theoretical expectations from projectile motion. Fitting the derived form  $d = 2\sqrt{H}\sqrt{h}$  enabled estimation of table height  $H$ , consistent with Galileo’s notes within uncertainty.

In Exercise 3, Ptolemy’s quadratic refraction model fit the data perfectly but likely reflected adjusted values rather than independent measurements. Snell’s sine-law fit yielded a refractive index close to the modern value and a borderline  $p$ -value, suggesting



either slight systematic errors or underestimated uncertainties.

## 5 Conclusion

In conclusion, the lab exercises demonstrated the power of least-squares fitting both as an analytical tool and as a means of testing theoretical models. We saw that by examining chi-squared and  $p$ -values, we can assess which model is statistically preferred for a given dataset. We also saw how important it is to consider the context of the data, including potential biases and the domain of applicability, rather than blindly trusting a mathematical fit. When used carefully, least-squares analysis not only yields best fit parameters but also deepens our understanding of the underlying physics.

## A Python Code

```
# Exercise 1
import numpy as np
from scipy.optimize import curve_fit
from scipy.stats import chi2
import matplotlib.pyplot as plt

x_data = np.array([1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0])
y_data = np.array([2.7, 3.9, 5.5, 5.8, 6.5, 6.3, 7.7, 8.5, 8.7])
sigma_data = np.array([0.3, 0.5, 0.7, 0.6, 0.4, 0.3, 0.7, 0.8, 0.5])

def poly_func(x, *theta):
    return sum(theta[i] * x**i for i in range(len(theta)))

e1_results = []

for M in [1, 2, 3]:
    p0 = np.ones(M + 1)

    theta_hat, cov = curve_fit(lambda xx, *params: poly_func(xx, *params), x_data, y_data,
                               p0=p0, sigma=sigma_data, absolute_sigma=True)

    y_fit = poly_func(x_data, *theta_hat)
    chi_sq = np.sum(((y_data - y_fit) / sigma_data) ** 2)
    ndof = len(x_data) - len(theta_hat)
    p_val = 1 - chi2.cdf(chi_sq, ndof)

    e1_results.append((M, theta_hat, cov, chi_sq, ndof, p_val))

    print(f"Polynomial order M={M}")
    print(f" Parameters: {theta_hat}")
    print(f" Parameter errors: {np.sqrt(np.diag(cov))}")
    print(f" chi^2 = {chi_sq:.3f}, ndof = {ndof}, p-value = {p_val:.3f}\n")

x_grid = np.linspace(0, 20, 200) # grid for plotting
for M, theta_hat, cov, chi_sq, ndof, p_val in e1_results:

    f_vals = poly_func(x_grid, *theta_hat)

    design_matrix = np.vstack([x_grid ** i for i in range(len(theta_hat))]).T
    sigma_f2 = np.einsum('ij,jk,ik->i', design_matrix, cov, design_matrix)
    sigma_f = np.sqrt(sigma_f2)

    plt.figure()
    plt.errorbar(x_data, y_data, yerr=sigma_data, fmt='o', label='data')
    plt.plot(x_grid, f_vals, label=f'fit M={M}')
    plt.fill_between(x_grid, f_vals - sigma_f, f_vals + sigma_f, alpha=0.3,
                     label=f'±1 band')
```



```

plt.title(f'Polynomial fit M={M}')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.show()
plt.savefig(f'ex1_poly_M{M}.png')
plt.close()

M3_theta_hat = e1_results[-1][1]
M3_cov = e1_results[-1][2]

def delta_ab(a, b, theta, cov):

    f_a = poly_func(a, *theta)
    f_b = poly_func(b, *theta)
    delta_val = f_a - f_b

    terms_a = np.array([a ** i for i in range(len(theta))])
    terms_b = np.array([b ** i for i in range(len(theta))])
    dvec = terms_a - terms_b
    var_delta = dvec @ cov @ dvec
    sigma_delta = np.sqrt(var_delta)
    return delta_val, sigma_delta

for b_val in [6, 10, 20]:
    delta_val, sigma_delta = delta_ab(5, b_val, M3_theta_hat, M3_cov)
    print(f"Delta(f(5) - f({b_val})) = {delta_val:.3f} ± {sigma_delta:.3f}")

# Exercise 2
h_data = np.array([1000, 828, 800, 600, 300])
d_data = np.array([1500, 1340, 1328, 1172, 800])
sigma_d = np.full_like(h_data, 15)

def model_linear(h, alpha):
    return alpha * h

def model_quadratic(h, alpha, beta):
    return alpha * h + beta * h**2

def model_powerlaw(h, alpha, beta):
    return alpha * h**beta

def model_sqrt(h, A):
    return A * np.sqrt(h)

model_specs = [
    ('Linear', model_linear, [1.0]),
    ('Quadratic', model_quadratic, [2.0, -1e-3]),
    ('Power-law', model_powerlaw, [40, 0.5])
]

e2_results = []

for name, func, p0 in model_specs:
    popt, pcov = curve_fit(func, h_data, d_data, p0=p0, sigma=sigma_d,
                           absolute_sigma=True, maxfev=10000)
    resid = d_data - func(h_data, *popt)
    chi_sq = np.sum((resid / sigma_d) ** 2)
    ndof = len(h_data) - len(popt)
    p_val = 1 - chi2.cdf(chi_sq, ndof)
    perr = np.sqrt(np.diag(pcov))
    corr_mat = None
    if len(popt) > 1:

```

```

    corr_mat = pcov / np.outer(perr, perr)
    e2_results.append((name, popt, perr, pcov, corr_mat, chi_sq, ndof, p_val))

    print(f"\n{name} model fit:")
    print(f"  Parameters: {popt}")
    print(f"  Errors: {perr}")
    print(f"   $\chi^2 = \{chi\_sq:.3f\}$ , ndof = {ndof}, p-value = {p_val:.3f}")
    if corr_mat is not None:
        print(f"  Correlation matrix:\n{corr_mat}")

sqrt_popt, sqrt_pcov = curve_fit(model_sqrt, h_data, d_data, p0=[10],
                                  sigma=sigma_d, absolute_sigma=True)
A_val = sqrt_popt[0]
A_err = np.sqrt(np.diag(sqrt_pcov))[0]
chi_sq_sqrt = np.sum(((d_data - model_sqrt(h_data, *sqrt_popt)) / sigma_d) ** 2)
ndof_sqrt = len(h_data) - len(sqrt_popt)
p_val_sqrt = 1 - chi2.cdf(chi_sq_sqrt, ndof_sqrt)

H_est = (A_val / 2)**2
H_err = (A_val / 2) * A_err
print("\nSqrt model fit:")
print(f"  A = {A_val:.3f} ± {A_err:.3f}")
print(f"   $\chi^2 = \{chi\_sq\_sqrt:.3f\}$ , ndof = {ndof_sqrt}, p-value = {p_val_sqrt:.3f}")
print(f"  Estimated table height H = {H_est:.3f} ± {H_err:.3f}\n")

# Exercise 3
theta_i_deg = np.array([10, 20, 30, 40, 50, 60, 70, 80])
theta_r_deg = np.array([8.0, 15.5, 22.5, 29.0, 35.0, 40.5, 45.5, 50.0])
sigma_deg = np.full_like(theta_i_deg, 0.5) # assume 0.5 degree uncertainty

def model_pt_linear(theta_i, alpha):
    return alpha * theta_i

def model_pt_quadratic(theta_i, alpha, beta):
    return alpha * theta_i - beta * theta_i ** 2

def model_pt_snell(theta_i, r):

    theta_i_rad = np.deg2rad(theta_i)
    theta_r_rad = np.arcsin(np.sin(theta_i_rad) / r)
    return np.rad2deg(theta_r_rad)

for name, func, p0 in [
    ('Linear', model_pt_linear, [0.6]),
    ('Quadratic', model_pt_quadratic, [0.8, 0.0])
]:
    popt, pcov = curve_fit(func, theta_i_deg, theta_r_deg, p0=p0,
                          sigma=sigma_deg, absolute_sigma=True)
    y_fit = func(theta_i_deg, *popt)
    chi_sq = np.sum(((theta_r_deg - y_fit) / sigma_deg) ** 2)
    ndof = len(theta_i_deg) - len(popt)
    p_val = 1 - chi2.cdf(chi_sq, ndof)
    perr = np.sqrt(np.diag(pcov))
    corr = None
    if len(popt) > 1:
        corr = pcov / np.outer(perr, perr)
    print(f"\n{name} model fit for Ptolemy data:")
    print(f"  Parameters: {popt}")
    print(f"  Errors: {perr}")
    print(f"   $\chi^2 = \{chi\_sq:.3f\}$ , ndof = {ndof}, p-value = {p_val:.3f}")
    if corr is not None:
        print(f"  Correlation matrix:\n{corr}")

```

```

snell_popt, snell_pcov = curve_fit(model_pt_snell, theta_i_deg, theta_r_deg, p0=[1.33],
                                sigma=sigma_deg, absolute_sigma=True)
theta_fit_snell = model_pt_snell(theta_i_deg, *snell_popt)
chi_sq_snell = np.sum(((theta_r_deg - theta_fit_snell) / sigma_deg) ** 2)
ndof_snell = len(theta_i_deg) - len(snell_popt)
p_val_snell = 1 - chi2.cdf(chi_sq_snell, ndof_snell)

print("\nSnell's law fit for Ptolemy data:")
print(f"  r = {snell_popt[0]:.4f} ± {np.sqrt(np.diag(snell_pcov))[0]:.4f}")
print(f"  chi^2 = {chi_sq_snell:.3f}, ndof = {ndof_snell}, p-value = {p_val_snell:.3f}")

```

## References

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4. Appendix A of PH3010 lab materials: Galileo's original table height (punti 828) extracted from handwritten manuscript.
5. Rhodes, J. (2006). *Ptolemy's Refraction Data and the Development of Snell's Law*, in *European Journal of Physics*, 27(5), 1037–1050.